

Mohamed Essam Madyan

Full-Stack Developer – Next.js & Node.js | MERN Stack

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OBJECTIVE

Full-Stack Developer specializing in Next.js and Node.js with hands-on experience building RESTful APIs and full-stack web applications using MongoDB, PostgreSQL with Prisma ORM and JWT authentication. Delivered 5 complete projects including an AI-powered application (Expert AI), a Social Media API, a POS System, and a graduation-level service marketplace (San3a). Trained at ITI and NTI with a BSc in Computer Science from Minufiya University.

EDUCATION

BSc in Computer Science – Software Engineering

Minufiya University, Faculty of Computers and Information | 2020 – 2024

PROFESSIONAL EXPERIENCE

NTI – National Telecommunication Institute Full-Stack Developer Trainee | Minufiya | 2026

- Developed full-stack web applications using React.js and Next.js (SSR/SSG) with server-side rendering to improve page load performance
- Built secure RESTful APIs using Node.js and Express.js with JWT-based authentication and role-based access control (RBAC) protecting 10+ endpoints
- Designed and optimized MongoDB schemas using Mongoose with data validation, indexing, and population for complex relational queries
- Containerized full-stack applications using Docker and automated build/deploy pipelines with GitHub Actions reducing manual deployment steps
- Deployed full-stack apps to AWS, Vercel, Netlify with automated testing (Jest, Cypress)
- Built capstone platform with real-time notifications via Socket.io and integrated Cloudinary for image uploads with cloud storage management

ITI – Information Technology Institute Front-End Developer Trainee | Minufiya | 2024

- Built responsive user interfaces using React.js, TypeScript, and Tailwind CSS
- Implemented state management with Context API and React Hooks
- Developed reusable component architecture and responsive layouts with Flexbox/Grid
- Integrated RESTful APIs using Fetch API and Axios for dynamic data rendering

Route IT Training Center Front-End Developer Trainee | Cairo | 2024

- Built responsive user interfaces with HTML5, CSS3, JavaScript (ES6+), and TypeScript
- Developed interactive web pages with DOM manipulation and event handling
- Implemented responsive layouts using Flexbox, Grid, and media queries
- Styled components with Tailwind CSS and Bootstrap following modern design principles

TECHNICAL SKILLS

Front-End : Next.js, React.js, TypeScript, Tailwind CSS, Redux

Back-End : Node.js, Express.js, RESTful APIs, MVC Architecture, Authentication & Authorization (JWT, OAuth2), WebSockets (Socket.io), bcrypt, Joi, helmet, cors, morgan, dotenv

Databases: MongoDB, PostgreSQL, Mongoose, Prisma ORM

DevOps & Tools: Docker, GitHub Actions, AWS, Vercel, Git, Linux, Postman

CERTIFICATES & TRAINING

Hire Ready Program – Open Source Track, NTI (4 months) | 2026

Front-End Diploma– ITI, Information Technology Institute | 2024

Front-End Diploma– Route IT Training Center | 2024

Back-End Diploma – The Complete Node.js Bootcamp, Udemy (Jonas Schmedtmann) | 2024

PROJECTS

Expert AI – Full-Stack AI-Powered Application

Stack: Node.js, Express.js, React.js, MongoDB, Groq API, TypeScript

- Built a full-stack AI-powered web application featuring an intelligent chatbot, portfolio showcase, and smart navigation using Node.js, Express.js, and React.js
- Integrated Groq API to enable fast AI-driven conversational responses with context-aware chat functionality
- Architected the backend using MVC pattern with dedicated Controllers, Middlewares, Routes, and Models for clean and scalable code structure
- Implemented JWT authentication and custom middleware for request validation and error handling
- Managed environment configuration using dotenv with a documented .env.example for clean project setup

Social Media API - Backend Project

- Architected a RESTful Social Media API featuring user profiles, posts, comments, likes, and follow/unfollow relationships using Node.js, Express.js, and MongoDB with Mongoose
- Implemented JWT-based authentication with protected routes, middleware-level authorization, and centralized error handling with proper HTTP status codes using express-validator
- Documented and tested 20+ API endpoints using Postman collections, covering authentication flows, CRUD operations, and edge case error responses

POS System – Full Stack Project

- Built a full-featured Point-of-Sale system handling product inventory, customer management, and invoice generation with real-time data updates
- Developed RESTful APIs for inventory and transaction management using Node.js and Express.js with MongoDB and Mongoose for data modeling
- Created responsive React front-end with clean UI and real-time data updates.

San3a – Full-Stack Web Application (Graduation Project)

- Built a service marketplace platform using Next.js and Node.js enabling clients to search, filter, and book verified craftsmen with role-based dashboards for each user type
- Designed and implemented scalable RESTful APIs and managed MongoDB schemas for diverse user roles (Clients & Craftsmen).
- Built a fully responsive UI using Next.js with dynamic routing and optimized page rendering across all devices
- Integrated real-time booking notifications using Socket.io  to alert craftsmen of new requests and clients of status updates

Mayorca Travel — Pilgrim Management & Document Library ERP

- Built a full-stack ERP system for managing Hajj and Umrah pilgrim operations using React.js and Node.js with role-based access control (Admin/Employee) and granular permission management for each user type
- Designed and implemented scalable RESTful APIs with Express.js and MongoDB, handling complex data models for pilgrims, delegates, trips, contracts, flight routes, and document management across the entire pilgrimage workflow
- Built a fully responsive dashboard UI using React with lazy-loaded routes, dynamic routing, and Tailwind CSS for optimized performance across all devices
- Integrated real-time activity monitoring with device fingerprinting (FingerprintJS) and session management, plus secure file uploads with Sharp image processing and Excel export functionality for operational reports

SKILLS & ABILITIES

- Strong problem-solving skills with an analytical mindset.
- Effective communicator and collaborative team player.
- Quick learner with strong adaptability to new technologies.
- Basic knowledge of media buying and digital marketing concepts.
- Entrepreneurial mindset with strong business awareness
- Effective time management and task prioritization skills.

LANGUAGES

Arabic
Native

English
Advanced

MILITARY STATUS

Completed (2025) – Awarded Certificate of Good Conduct